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13 February 1979
SUPERSEDING
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21 April 1975

MILITARY SPECIFICATION

CLOTH, TWILL, POLYESTER/COTTON (DURABLE PRESS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers blended polyester and cotton cloth, chemically treated with a durable press finish, for manufacturing durable press uniforms.

* 1.2 Classification. The cloth covered by this specification shall be of the following types, as specified (see 6.2):

Type I for dress uniforms (postcured resin application)

Type II for utility uniforms

Class 1 - Postcured resin application
Class 2 - Precured resin application
Class 3 - Precured resin application containing soil release resins

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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SPECIFICATIONS

FEDERAL

PPP-P-1134 - Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

MILITARY

MIL-T-43548 - Thread, Polyester, Cotton-Covered
MIL-T-43624 - Thread, Polyester, Spun

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

RULES AND REGULATIONS UNDER THE TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

TECHNICAL MANUAL OF THE AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

Method No. 119-1970 Color Change Due to Flat Abrasion (Frosting): Screen Wire Method

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Method No. 124-1978 Appearance of Durable Press Fabrics after Repeated Home Launderings

Method No. 88C-1975 Appearance of Creases in Wash and Wear Items after Home Laundering

(Application for copies should be addressed to AATCC National Headquarters, P. O. Box 12215, Research Triangle Park, NC 27709.)

3. REQUIREMENTS

3.1 Standard sample. The dyed, treated, pressed and cured cloth, shall match the standard sample (which has been pressed and cured for shade), and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Materials (see 6.7).

3.2.1 Fiber blend.

* 3.2.1.1 Polyester fiber. The fiber shall be polyethylene glycol terephthalate, semi-dull. The denier shall not be greater than 2.5 (2.8 dtex) and the staple shall be approximately 1-1/2 inch (38 mm) in length.

3.2.1.2 Cotton fiber. Type I - The cotton shall be combed.
Type II - The cotton shall be carded or combed.

3.2.2 Yarn. The warp and filling yarn shall be a blend of 50 ± 5 percent polyester and the remaining percentage cotton, based on the dry weight of the dyed cloth before treatment, when tested as specified in 4.4. The warp shall be a 2-ply yarn for type I and singles for type II and the filling shall be singles yarn for both types (see 6.3.1).

3.3 Color. Unless otherwise specified, the color of the cured type I cloth shall be Army Tan 445 to match the standard sample and the color of the cured type II cloth shall be Olive Green 507 to match the standard sample (see 6.3). The color for type I cloth shall be imparted by the use of selected vat dyes or a combination of selected vat and dispersed dyes. The color for type II cloth shall be imparted by the use of a combination of selected vat and dispersed dyes. The color for both types I and II cloth shall be applied by a continuous thermal fixation, steam reduction and oxidation procedure (see 6.3.2).

* 3.3.1 Labile sulfur. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyed pressed and cured cloth shall show not more than a slight trace of labile sulfur as defined in the test method, when tested as specified in 4.4.

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* 3.3.2 Matching. The color of the cured cloth shall match the standard sample under artificial daylight having a color temperature of 7500 kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800 kelvin, when examined as specified in 4.2.2.3 (see 6.3.2).

* 3.3.3 Colorfastness. The cured cloth shall show fastness to laundering (after 3 cycles) crocking, perspiration, frosting, sublimation and light equal to or better than the standard sample when tested as specified in 4.4. When no standard sample is available, or when the standard sample is not referenced for colorfastness, the cured cloth shall show "good" colorfastness to laundering (after 3 cycles), perspiration, sublimation, light after 40 hours and no more crocking than Munsell Value 8.5, also no more frosting than 4-5 on the grey scale, when tested as specified in 4.4.

* 3.4 Physical requirements. The cured cloth shall conform to the requirements specified in table I when tested as specified in 4.4.

TABLE I. Physical requirements

Type	Weight oz/sq. yd. (g/m^2)	Yarns per inch (per cm) minimum		Breaking strength lbs. (Newtons) minimum		Tearing strength lbs. (Newtons) minimum	
		Warp	Filling	Warp	Filling	Warp	Filling
I	7.3 ± 0.5 (247 ± 17)	104 (41)	50 (20)	170 (756)	80 (356)	5.0 (22)	3.0 (13)
II	7.3 ± 0.5 (247 ± 17)	85 (33)	45 (18)	145 (645)	90 (400)	5.0 (22)	5.0 (22)

3.4.1 Width. The width of the cloth shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of selvages.

3.4.2 Weave. Type I - The weave shall be 2 up - 1 down right hand twill.
Type II - The weave shall be 2 up - 1 down left hand twill.

3.5 Cloth preparation. The type I cloth shall be desized, scoured, mercerized, bleached and heat set prior to dyeing. The type II cloth shall be desized, scoured, mercerized and heat set prior to dyeing.

3.6 Nonfibrous materials. The starch and protein content including chloroform-soluble and water-soluble material of the scoured and dyed cloth shall not exceed 2.0 percent when tested as specified in 4.4.

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3.7 Dimensional stability.

* 3.7.1 (For Government procurements of type I and type II, class 1 cloth only). The resin or reactant treated dyed cloth shall not shrink more than 2.0 percent in the warp and 1.5 percent in the filling under the curing conditions specified in footnote 2/ of table III, when tested as specified in 4.4 (see 6.6).

3.7.2 The cured cloth shall have dimensional changes of not more than 2.5 percent in the warp and 1.5 percent in the filling after 15 launderings when tested as specified in 4.4.

3.8 Treatment.

* 3.8.1 Type I and type II, class 1. The dyed cloth shall be given an approved (see 6.4), durable press and soil release treatment, consisting of a suitable catalyzed resin or reactant system in combination with a soil release agent and hand modifiers. A wetting agent may be used to facilitate processing. Durable press resins and soil release agents other than those approved for use are prohibited. The cloth shall be processed so that the uncured and cured cloth shall meet the applicable requirements of this specification when tested as specified in 4.4.

* 3.8.2 Type II, class 2. The dyed cloth shall be given an approved (see 6.4), durable press treatment. Durable press resins other than those approved for use are prohibited. The cloth shall be processed so that the cured cloth shall meet the applicable requirements when tested as specified in 4.4.

* 3.8.3 Type II, class 3. The dyed cloth shall be treated as specified in 3.8.1. The precured treated cloth shall meet the applicable requirements of this specification when test as specified in 4.4.

* 3.8.4 Appearance, soil release and crease retention. The cured cloth shall show ratings not less than the following when tested after 15 launderings as specified in 4.4.

	Minimum average ratings per sample unit
Appearance rating	4.0
Soil release rating (not applicable for class 2)	3.0
Crease retention rating (not applicable for classes 2 and 3)	4.0

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3.8.5 Spray rating limit. The results of the three individual determinations for spray rating on the cured cloth shall not be greater than the maximum values 70, 70, 70 when tested following 3 launderings as specified in 4.4.

3.9 Seam efficiency.

* 3.9.1 Type I and type II, class 1. The dyed, treated and uncured cloth shall have a seam efficiency of not less than 90 percent when tested as specified in 4.4.

* 3.9.2 Type II, classes 2 and 3. The dyed treated and cured cloth shall have a seam efficiency of not less than 80 percent when tested as specified in 4.4.

* 3.10 pH. The pH value of the water extract of the cured cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.4.

* 3.11 Length and put-up (for Government procurement cloth only). Unless otherwise specified (see 6.2), the treated and uncured cloth for type I and type II, class 1, and the cured cloth for type II, classes 2 and 3 shall be furnished in continuous lengths each not less than 40 yards. Each length shall be put up on rolls as specified in PPP-P-1134.

* 3.12 Fiber identification. Each roll shall be labeled or ticketed for fiber content in accordance with the Textile Fiber Products Identification Act.

3.13 Face identification. The face side of the cloth shall be identified by stamping on that side with the word "Face" at each end of the roll.

3.14 The piece tickets for the cloth (see 5.3), shall show the date of application of the durable press and soil release treatment by month and year or by actual date.

3.15 Workmanship. The cloth shall conform to the quality established by this specification. The demerit points per 100 square yards (84 m^2) when calculated as specified in section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performace of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.2.2 Examination of the end item. Examination of the end item shall be in accordance with 4.2.2.1 through 4.2.2.4.

4.2.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards (91 m), the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards (91 m), only 100 yards (91 m) shall be examined. All defects as defined in Section I of FED-STD-4, which are clearly noticeable at normal inspection distance 3 feet (1 m) shall be scored and assigned demerit points as listed in 4.2.2.1.1, except that only those slubs and knots which exceed the limits shown on Sears Fabric Defect Scales (6.5), D for slubs and B for knots, shall be scored. No linear yard (increments of 1 yard (0.91 m) on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (rolls) 1/
3200 or less	8
3201 up to and including 10,000	13
10,001 and over	20

1/ No more than one roll shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of rolls in which case all shipping containers shall be present in the sample.

The lot shall be unacceptable if the points per 100 square yards (84 m^2) of the total yardage examined exceeds 30 points. The lot shall be unacceptable if the points per 100 square yards (84 m^2) of two or more individual rolls exceeds 45 points. If one roll exceeds 45 points per 100 square yards (84 m^2), a second sample equal in size to that of the first sample shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 45 points per 100 square yards (84 m^2). Point computation for lot quality and individual roll quality shall be as follows:

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$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches (76 mm) or less in any dimension - one point

For defects exceeding 3 inches (76 mm), but not exceeding 6 inches (152 mm) in any dimension - two points

For defects exceeding 6 inches (152 mm), but not exceeding 9 inches (229 mm) in any dimension - three points

For defects exceeding 9 inches (229 mm) in any dimension - four points

The following defects, when present, shall be scored four points for each yard (0.91 m) in which they occur:

Baggy, ridgy or wavy cloth
Width less than specified
Poor dye penetration, mottled, streaky or cloudy
Overall uncleanness
Excessive neppiness

4.2.2.2 Examination for length.

4.2.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards (1.8 m) less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.2.2.3 Examination for shade. Each roll in the lot shall be examined visually for shade (see 3.3.2). The sample unit for examination shall be a piece not less than 5 inches by 8 inches (127 mm by 203 mm) pressed and cured. A sample unit shall be drawn from each roll in the lot and identified with the roll number. A roll shall be unacceptable if the sample unit fails to meet the requirement for shade matching. When specified (see 6.2), a second piece shall also be taken and made available for reference in the uncured condition.

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4.2.2.4 Examination for face identification and compliance with Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for the defects listed below:

Face identification missing from either or both ends.

Face identification on wrong side.

Not labeled in accordance with the Textile Fiber Products Identification Act.

Each roll not labeled in accordance with this act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

4.3 Packaging inspection. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that preservation, packing and marking comply with the section 5 requirements.

4.3.1 Examination for dating of treatment. An examination shall be made to determine that the piece ticket shows the date of the application of the durable press and soil release treatment. Each roll not containing this information on the piece ticket shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

4.4 Testing of the end tem. The methods of testing specified in FED-STD-191, where applicable, and as listed in tables II and III, shall be followed. The physical and chemical values specified in section 3, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values as well as the final result. The sample unit for testing as specified in table II shall be 1/2 yard (0.46 m) of the dyed cloth, prior to application of the treatment. For the uncured cloth, tests in table III, the sample unit will be 3-1/2 yards (3.2 m) of dyed, treated and uncured cloth. For the cured cloth tests, the sample unit shall be 3-1/4 yards (3.0 m), full width, of dyed treated and cured cloth. The 3-1/4 yard (3.0 m) unit may be cut into 3 approximately equal parts to facilitate curing, provided each piece is marked to retain the identity of the sample unit. The sample size (number of sample units) shall be as shown in the table below. The lot size shall be expressed in units of 1 yard (0.91 m). The lot shall be unaceptable if one or more sample units fail to meet any test requirements specified.

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Sample size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II. Test methods (applicable to dyed cloth, prior to treatment)

Characteristic	Requirement paragraph	Test method
Polyester identification	3.2.1.1	1/
Cotton identification	3.2.1.2	1/
Polyester fiber content:		
Warp yarn	3.2.2	4.4.1 1/
Filling yarn	3.2.2	4.4.1 1/
Yarn ply	3.2.2	Visual 2/
Desized	3.5	1/
Scouring	3.5	1/
Mercerized	3.5	1/
Bleached (Type I)	3.5	1/
Heat set	3.5	1/
Weave	3.4.2	Visual 2/
Nonfibrous materials	3.6	2611

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ One determination for each sample unit and the result reported as "pass" or "fail".

TABLE III. Test methods (applicable to treated cloth)

Characteristic	Requirement paragraph	Test method
Dimensional stability during curing (Type I and Type II, class 1)	3.7.1	2/
Cured cloth:		
Labile sulfur	3.3.1	2020
Colorfastness to:		
Laundering (after 3 cycles)	3.3.3	5610
Crocking	3.3.3	5651

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TABLE III. Test methods (applicable to treated cloth) (cont'd)

Characteristic	Requirement paragraph	Test method
Cured cloth (cont'd)		
Colorfastness to: (cont'd)		
Perspiration	3.3.3	5680
Light	3.3.3	5660
Frosting	3.3.3	4.4.2 4/
Sublimation	3.3.3	5642 3/
Weight	3.4	5041
pH	3.10	2811
Yarns per inch:		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength:		
Warp	3.4	5100
Filling	3.4	5100
Tearing strength:		
Warp	3.4	5132
Filling	3.4	5132
Dimensional stability after 15 launderings:		
Warp	3.7.2	4.4.3
Filling	3.7.2	4.4.3
Appearance after 15 launderings	3.8.4	4.4.3
Crease retention after 15 launderings	3.8.4	4.4.3
Soil release after 15 launderings	3.8.4	4.4.3
Spray rating		
After 3 launderings	3.8.5	4.4.3.2 and 5526
Seam efficiency	3.9	5110 1/

1/ The needle shall measure 0.044 ± 0.001 inch (1.12 ± 0.3 mm) across the blade at the eye. The thread shall be either polyester/cotton in accordance with MIL-T-43548, ticket No. 50, 2 ply for the needle and ticket No. 70, 2 ply for the looper or spun polyester in accordance with MIL-T-43624, class 1 or class 2 ticket No. 40, 3 ply for the needle and ticket No. 70, 2 ply for the looper.

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2/ Three specimens 22 by 22 inches (559 by 559 mm) for testing dimensional stability during curing shall be cut from treated but uncured cloth, and be marked as described in 4.4.3.2.1. Pressing and curing shall be as recommended by the fabric supplier. Lacking specific instructions the following procedure shall be used: The specimens shall be pressed on a hot head press at 310°F (154°C) and 80 pounds per square inch (552 kPa) gauge pressure under the following cycle at full pressure, and then conditioned and remeasured as described in 4.4.3.2.

5 seconds steam
5 seconds bake
5 seconds vacuum during bake
release head and continue vacuum
for 5 seconds.

Curing cycle: Pressed fabric should be cured at a minimum of 325°F (163°C) for a period of 12 minutes.

3/ The test shall be conducted at a temperature range of 5.

4/ The frosting test shall be conducted on cured cloth after 15 cycles of laundering as specified in 4.4.3.2.2.

4.4.1 Polyester fiber content.

4.4.1.1 Test specimen. The specimen shall be approximately 2 grams made up of yarns ravelled from the dyed cloth sample and cut into approximately 1/4 inch (6 mm) lengths.

4.4.1.2 Apparatus, reagents, method cited.

Apparatus as described in method 2611.

Sintered glass filter crucible of coarse porosity.

Hydrometer.

Sulfuric acid, 75 percent (density 1.67) prepared by pouring first 350 ml of water in a round-bottom flask, then adding 680 ml of concentrated sulfuric acid slowly, with cooling under the tap. The density of the solution is adjusted to 1.67 using a hydrometer.

Sulfuric acid, dilute (100 ml of concentrated sulfuric acid added to 1900 ml of water).

4.4.1.3 Procedure. The nonfibrous material shall be removed as specified in Method 2611. The desized specimen shall be dried to a constant weight, cooled in a desiccator and weighed. The specimen shall then be immersed in

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200 ml of 75 percent solution of sulfuric acid for 1 hour with intermittent stirring. The residual fibers shall be transferred to the glass crucible and the excess acid solution removed by suction. The residual fibers shall be washed on the fiber filter successively with dilute sulfuric acid, water, dilute ammonia solution (soak for ten minutes) and again with water (soak for ten minutes). After each separate washing, the crucible is drained by suction. The residue shall be dried to a constant weight in the air oven at 105° to 110°C. This is the "weight of the dry polyester fiber" in the specimen.

4.4.1.4 Result. The percentage of polyester fiber shall be based on the following equation:

$$\text{Polyester fiber, percent} = \frac{\text{Weight of dry polyester fiber}}{\text{Weight of dry desized specimen}} \times 100$$

4.4.1.5 Two specimens shall be tested from each sample unit and the polyester fiber content shall be the average of the results obtained from the specimen tested and shall be reported to the nearest 1.0 percent.

4.4.2 Frosting. The test method used shall be AATCC Test Method 119-1970 Color Change Due to Flat Abrasion (Frosting): Screen Wire Method.

4.4.3 Appearance of cloth, crease retention, soil release and the determination of dimensional stability.

4.4.3.1 Apparatus and material.

4.4.3.1.1 The following apparatus and materials are required:

(a) Kenmore Automatic Washing Machine Model 600 or similar machine (in case of disputes, the Kenmore Model 600 shall be used).
(b) Kenmore Automatic Dryer Model 600 or equivalent.
(c) AATCC Standard Detergent 124 or equivalent.
(d) Mineral Oil; Nujol, Plough Inc., Memphis, Tennessee or equivalent.
(e) AATCC Standard Blotting paper or equivalent.
(f) Five pound cylindrical weight, 2-1/2 inches (64 mm) diameter.
(g) Dummy pieces for ballast - 36 x 36 inches (914 x 914 mm) hemmed of fabric similar to test specimens added to test specimens to make an approximately 4-pound (1.8 kg) load.
(h) AATCC Glassine paper or equivalent.

4.4.3.1.2 The following apparatus and materials are required for evaluation of samples:

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(a) Overhead lighting procedure. Apparatus shall be as specified in AATCC Test Method 124-1978, Appearance of Durable Press Fabrics after Repeated Home Launderings. In addition AATCC three dimensional Durable Press Replicas, Deering Milliken Corporation photograph standards for evaluating stain release and a non-reflecting black topped table 35 ± 1 inch (889 ± 25 mm) high shall also be required. The dimensions of the table shall be at least 30 x 24 inches (762 x 610 mm) with the long dimension parallel to the viewing board.

(b) Overhead and side lighting. Apparatus shall be as specified in AATCC Test Method 88C-1975, Appearance of Creases in Wash and Wear Items after Home Laundering. In addition photographic standards for evaluating pressed-in creases shall be required.

4.4.3.1.3 Evaluation shall be conducted in a room where the only source of light will be that used for observation. To prevent unwanted reflection of light that will interfere with the observation, the walls of the room may be painted a flat non-reflecting black or blackout curtains may be hung on each side of the viewing area.

4.4.3.2 Tests for characteristics involving laundering.

4.4.3.2.1 Preparation of specimens. Three specimens 22 x 22 inches (559 x 559 mm) measured parallel to the warp and filling shall be cut from a portion of the treated cloth sample which has been pressed and cured flat. One specimen shall be cut from each side of the sample unit to within three inches of the selvages and the third specimen shall be taken from the center. No two specimens shall contain the same filling yarns. The specimens shall be conditioned to equilibrium under Standard Conditions in accordance with FED-STD-191. The conditioned specimens shall be laid without tension on a flat surface, care being taken that the cloth is free from wrinkles or creases. Three distances, each a minimum of 18 inches (457 mm), shall be measured and marked off parallel to each of the warp and filling directions of the specimen. Each pair of markings shall be a minimum of 6 inches (152 mm) from each other and not closer than one inch to the edges of the specimen. The distance may be marked with indelible ink and a fine pointed pen, or by sewing fine threads into the cloth, or by stamping. The measured distance shall be parallel to the respective yarns. Three specimens 15 x 15 inches (381 x 381 mm) shall also be cut from an uncured portion of the treated cloth sample. These specimens shall be cured with a warpwise crease pressed through the center of the specimen. If specimens from these two sets are not to be used for spray rating tests following three launderings, a third set of specimens at least 8 x 8 inches (203 x 203 mm) shall be prepared. The two (or three) sets of samples shall then be laundered in accordance with 4.4.3.2.2.

* 4.4.3.2.2 Laundering. Place the three 15 by 15 inches (381 by 381 mm) creased specimens and the 22 by 22 inches (559 by 559 mm) specimens and the three 8 by 8 inches (203 by 203 mm) specimens for spray ratings in the washer. The washing long shall be 4 ± 1/4 pounds (1.8 ± 0.1 kg). Dummy pieces shall be added to the

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machine, along with the specimens to make up the 4 ± 1/4 pounds (1.8 ± 0.1 kg) load. Fill to the full water level of the washer with water of a hardness not to exceed 50 parts per million (ppm) and at a temperature of 115° ± 5°F (46 ± 3°C) and the rinse temperature shall be 80 ± 5°F (26 ± 3°C). Add 140 ± 1 grams of detergent. Set the washer for a 12 minute cycle on the "Normal" setting. Allow the washing to proceed automatically through the final spin cycle. Remove the specimens immediately at the completion of the final spin and separate from the dummy pieces and each other if tangled. Place the complete washed load (4 pounds) (1.8 kg) in the dryer and dry at the "Normal" cycle or a setting which generates an exhaust temperature range of 140° to 160°F (60° to 71°C). Operate the dryer until the load is dry and continue tumbling 5 minutes with the heat turned off (cool down cycle). Remove the load immediately after the machine stops. Repeat the wash and dry cycles for 3 complete cycles, following which the samples to be used for spray rating tests shall be removed from the load and conditioned for test. The load shall then be continued until 14 complete cycles. At the completion of the fourteenth cycle, the creased specimens to be used for the evaluation of the soil release characteristic shall be separated from any samples that are being tested for dimensional stability. Each specimen to be tested for soil release shall be stained with one spot of mineral oil. Place the area to be stained on AATCC blotting paper. Using an eyedropper stain the specimens using 5 drops to form the spot. Place a 3 x 3 inches (76 x 76 mm) piece of glassine paper directly over the oil. Place a 5 pound (2.3 kg) weight on the glassine paper. Keep the weight in place 60 seconds. Remove the weight and glassine paper and wipe off the excess oil with clean tissue. Allow the test specimen to hang undisturbed for at least 15 minutes but not over 2 hours. The stained specimens, any specimens that are being tested for dimensional stability and the dummy pieces required to make up the 4 pounds (1.8 kg) load are returned to the washing machine and laundered for the complete fifteenth wash and dry cycle. When the fifteenth cycle is completed, the specimens are conditioned to equilibrium under Standard Conditions in accordance with FED-STD-191. During conditioning and during periods of interruption of the cycles, the creased samples shall be suspended from a line with the crease running in the vertical direction to avoid distortion.

4.4.3.3 Evaluation. Three trained observers shall evaluate each sample unit for appearance, crease retention and soil release characteristics. The observers shall make their evaluations independent of each other. Each observer shall evaluate each specimen for appearance and crease retention as it hangs on the viewing board while standing in front of the viewing board and four feet back from it.

4.4.3.3.1 Dimensional stability. The laundered and conditioned specimens to be evaluated for dimensional stability shall be laid out without tension on a flat surface in the standard atmosphere, care shall be taken that the specimens are free from wrinkles and creases. The previously measured distances marked on the specimen shall be measured in both the warp and filling directions. The dimensional stability of the specimen shall be calculated as follows:

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$$\text{Dimensional change, percent} = \frac{A - B}{A} \times 100$$

Where: A = Average of initial measurements (3 specimens)
B = Average measurements after laundering (3 specimens)

The dimensional change of the sample unit in the warp and filling directions shall be the average of the specimens tested in each direction respectively and shall be reported separately to the nearest 0.1 percent. The individual values used to calculate the averages shall also be reported.

4.4.3.3.2 Evaluation of the appearance of the cloth. The overhead lighting shall be used (see 4.4.3.1.2a). Mount the creased specimen open to its full width on the viewing board with the center of the specimen five feet (1.5 m) from the floor with the crease in the vertical direction. Place three-dimensional plastic replicas on each side of the specimen with the centers five feet (1.5 m) from the floor, to facilitate comparative rating. The specimen shall be rated according to the appearance on the plastic replica that most nearly matches the appearance of the cloth. No estimated rating falling between the approved replicas shall be given. The average of the rating values assigned for appearance of the cloth by the observers rounded off to the nearest 0.1 rating shall be the rating for the specimen. Each rating value given by the observers for appearance of the cloth shall also be reported.

4.4.3.3.3 Evaluation of pressed-in creases. After evaluating the appearance of the cloth, the required light shall be added to the overhead lighting (see 4.4.3.1.2.b), and the specimen evaluated for the appearance of the pressed-in creases. Using the applicable standards, the specimens shall be compared with the standards and rated in the same manner as in 4.4.3.3.2. The average rating value rounded off to the nearest 0.1 rating for crease retention and the individual rating value given by each observer shall also be reported.

* 4.4.3.3.4 Evaluation of soil-release. The overhead lighting shall be used (see 4.4.3.1.2 a). The applicable standard shall be mounted on the viewing board 45 inches (1.1 m) from the floor. The cloth (shirt or trousers) shall be placed upon the black topped table with the stained area visible and in front of the standard. The observer shall stand 4 feet (1.2 m) in front of the viewing board. The stained area shall be compared with the standard and rated according to the nearest standard stain. No estimated ratings falling between standards are allowed. Any individual rating failing to meet the minimum requirement specified shall be cause for rejection. Rating values given by each observer shall be reported.

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* 5. PACKAGING

* 5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial as specified (see 6.2).

* 5.1.1 Levels A and Commercial. The cloth shall be put up and preserved in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

* 5.2.1 Levels A, B and Commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract and shipments shall be marked in accordance with the applicable requirements of PPP-P-1134 (see 3.14).

6. NOTES

* 6.1 Intended use. The type I cloth is intended for use in durable press dress uniforms. The type II, class 1 cloth is intended for use in durable press utility uniforms (fatigues). The type II, class 2 cloth is to be used as the base cloth for embroidered insignia, MIL-I-3037, and the type II, class 3 is to be used in the manufacture of Cap, Hot Weather, Olive Green 106, MIL-C-43724.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and class required (see 1.2).
(c) Width of cloth required (see 3.4.1).
(d) Length of roll required if other than specified (see 3.11).
(e) Whether uncured shade samples are required (see 4.2.2.3).
(f) Selection of applicable put-up, preservation, and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address procuring office issuing the invitation for bids.

* 6.3.1 The standard sample for type I cloth was woven from approximately 35/2 cotton count warp yarns and approximately 18/1 filling yarns. The requirements for type II cloth were based on fabrics having an approximate weight of 7.3 ounces per square yard (247 g/m^2) and woven with approximately 14/1 to 15.5/1 warp yarns and 12.5/1 to 14/1 filling yarns. The standard sample and subsequent production of type II fabric meeting this specification were prepared, dyed and finished with relaxed filling and minimum warp tension to insure maximum crimp interchange.

6.3.2 The standard sample for type I in Tan 445 was dyed by the thermal fixation process with the following dyestuff formulation:

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Disperse Blue 109
Disperse Blue 60
Disperse Orange 62
Disperse Red 140
Vat Brown 3 CI 69015/6
Vat Black 25 CI 69525
Vat Brown 1 CI 70800/1

The standard sample for type II in Olive Green 507 was dyed by the thermal fixation process with the following dyestuff formulation:

Vat Black 25 CI 69525
Vat Green 3 CI 69500/1
Vat Yellow P 2GA Pr. 717
Disperse Yellow 42 CI 10338
Disperse Blue 27 CI 60767
Disperse Red 72

Shade adjusting dyes may be used but the amount of such shade adjusting dyes should be held to an absolute minimum. Other formulations may be used at the discretion of the supplier provided that shade characteristics in both 7500 kelvin and 2800° kelvin light are comparable to that of the standard sample and all other color properties conform to the standard sample and such deviations in shade as may be permitted. Color matching under fluorescent tubes, even those that have been color corrected, is not recommended since many dyes on polyester fiber are adversely influenced by the nature of the radiation. Color matching on the part of the Government will be accomplished in accordance with Method 9010 of FED-STD-191 which uses the standard Macbeth skylight type source.

6.4 Approval of the durable press resins or reactants and soil release agents for utilization under this specification is the responsibility of the U.S. Army Natick Research and Development Command, Natick, MA 01760 and is based on extensive tests, including those for human toxicity which are not set forth in this specification. Because of the time necessary to conduct full evaluations, only those treatments approved and so listed in the invitation for bids or request for proposals shall be considered acceptable for the related procurement.

* 6.5 Sears Fabric Defect Scales are available from Sears, Roebuck and Company, Department 733Q, "Sears Defect Replica Kit", Sears Tower, Chicago, IL 60684.

6.6 In order to meet the requirements of 3.7.1, the dyed fabric had been sanforized prior to treating with a durable press finish.

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* 6.7 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification.

* 6.8 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

6.9 Marginal notations. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications corrections) from previous issues were made. This was done as a convenience only and Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notation and relationships to the last previous issue.

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Navy - SA
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Review activities:

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Preparing activity:

Army - GL

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-43791B - Cloth, Twill, Polyester/Cotton (Durable Press)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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